

## Scenario Topic Name: **Diabetes Charity Foundation**

### 1. Scenario and Example Queries

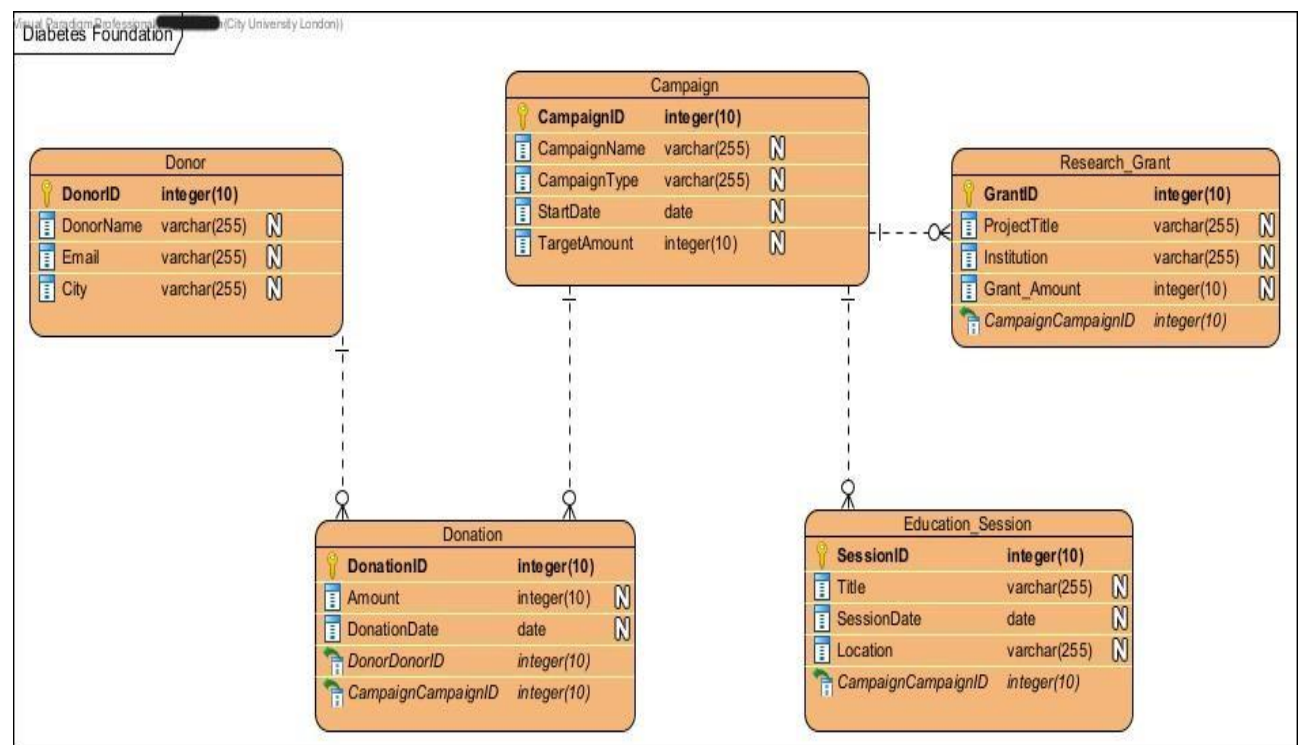
The Diabetes Charity Foundation is a medical and health charity that supports people living with diabetes. It runs fundraising campaigns, delivers patient education and awareness sessions, and raises funding for diabetes research projects.

Our charity needs a database to track donors, campaigns, donations, education sessions, and research grants. I will create the database to help the staff find answers to questions like: which donors give the most, which campaigns raise the highest amounts, how many educational sessions are delivered, and how campaign funds are used to support research. The database will help our charity organise its activities, and understand the impact of its work.

So, to make the queries or the questions clear, I will rewrite them as below:

1. Who are the donors who donated more than £500?
2. How many donations does each campaign receive?
3. Which education sessions will be organised under the campaign “Diabetes awareness week 2026”?
4. How many education sessions will happen in each location?
5. Which research projects receive grants above £10,000, and from which campaign?

### 2. Entity Relationship (ER) Model:



#### a. Entities and attributes:

1. Donor has attributes: DonorID (Primary Key), DonorName, Email, City.
2. Campaign has attributes: CampaignID (Primary Key), CampaignName, CampaignType (Fundraising, Awareness, Research), StartDate, TargetAmount.
3. Education Session has attributes: SessionID (Primary Key), Title, SessionDate, Location, CampaignID (Foreign Key from Campaign), this will be the event entity.
4. Donation has attributes: DonationID (Primary Key), DonorID (Foreign Key from Donor), CampaignID (Foreign Key from Campaign), Amount, DonationDate.
5. Research Grant has attributes: GrantID (Primary Key), ProjectTitle, Institution, Grant\_Amount, CampaignID (Foreign Key from Campaign).

#### b. The relationships:

1. A Donor can make many different Donations to the charity, but every Donation must come from one specific donor. Donor (one) TO (Many) Donation.
2. Each Campaign can also receive many Donations, but every Donation is linked to only one Campaign. Campaign (one) TO (Many) Donation.
3. A Campaign can organise several Education or awareness Sessions, but each session belongs to one Campaign only. Campaign (one) TO (Many) Education\_Session.
4. Campaigns can also fund Research Grants, one Campaign may support several Research Projects, but each Research Grant is connected to only one Campaign. Campaign (one) TO (Many) Research\_Grant

### 3. Relational Model Tables

#### a. Relational Model Tables

| Relational table specification                       |
|------------------------------------------------------|
| <b>Table name:</b> Donor                             |
| <b>Attributes</b>                                    |
| DonorID INT                                          |
| DonorName VARCHAR(100)                               |
| Email VARCHAR(100)                                   |
| City VARCHAR(50)                                     |
| <b>PRIMARY KEY</b> : DonorID                         |
| <b>FOREIGN KEY</b> : No foreign key for this entity. |

|                                                    |
|----------------------------------------------------|
| <b>Relational table specification</b>              |
| <b>Table name:</b> Campaign                        |
| <b>Attributes</b>                                  |
| CampaignID INT                                     |
| CampaignName VARCHAR(100)                          |
| CampaignType VARCHAR(100)                          |
| StartDate DATE                                     |
| TargetAmount INT                                   |
| <b>PRIMARY KEY</b> CampaignID                      |
| <b>FOREIGN KEY</b> No foreign key for this entity. |

|                                                                  |
|------------------------------------------------------------------|
| <b>Relational table specification</b>                            |
| <b>Table name:</b> Education_Session                             |
| <b>Attributes</b>                                                |
| SessionID INT                                                    |
| Title VARCHAR(100)                                               |
| SessionDate DATE                                                 |
| Location VARCHAR(100)                                            |
| CampaignID INT                                                   |
| <b>PRIMARY KEY</b> SessionID                                     |
| <b>FOREIGN KEY</b> (CampaignID) REFERENCES Campaign (CampaignID) |

|                                                                   |
|-------------------------------------------------------------------|
| <b>Relational table specification</b>                             |
| <b>Table name:</b> Donation                                       |
| <b>Attributes</b>                                                 |
| DonationID INT                                                    |
| Amount INT                                                        |
| DonationDate DATE                                                 |
| DonorID INT                                                       |
| CampaignID INT                                                    |
| <b>PRIMARY KEY</b> DonationID                                     |
| <b>FOREIGN KEY</b> (DonorID) REFERENCES Donor (DonorID)           |
| <b>FOREIGN KEY</b> ((CampaignID) REFERENCES Campaign (CampaignID) |

|                                                                  |
|------------------------------------------------------------------|
| <b>Relational table specification</b>                            |
| <b>Table name:</b> Research_Grant                                |
| <b>Attributes</b>                                                |
| GrantID INT                                                      |
| ProjectTitle VARCHAR(100)                                        |
| Institution VARCHAR(100)                                         |
| ApprovedAmount INT                                               |
| CampaignID INT                                                   |
| <b>PRIMARY KEY</b> GrantID                                       |
| <b>FOREIGN KEY</b> (CampaignID) REFERENCES Campaign (CampaignID) |

## b. Relational Model Tables with Values

|                                                                   |
|-------------------------------------------------------------------|
| <b>Relational table with Values specification</b>                 |
|                                                                   |
| <b>INSERT INTO Donor (DonorID, DonorName, Email, City) VALUES</b> |
| <b>(</b>                                                          |
| (1, 'Ani Bro', 'a.b@example.org', 'London'),                      |
| (2, 'Mo Kan', 'm.k@example.org', 'Birmingham'),                   |
| (3, 'Sar Le', 's.l@example.org', 'Manchester'),                   |
| (4, 'Jo Pat', 'j.p@example.org', 'Leeds'),                        |
| (5, 'Emi Cla', 'e.c@example.org', 'London')                       |
| <b>);</b>                                                         |

|                                                                                                       |
|-------------------------------------------------------------------------------------------------------|
| <b>Relational table with Values specification</b>                                                     |
|                                                                                                       |
| <b>INSERT INTO Campaign (CampaignID , CampaignName, CampaignType, StartDate, TargetAmount) VALUES</b> |
| <b>(</b>                                                                                              |
| (10, 'Diabetes Awareness Week 2026', 'Awareness', '2026-03-01', 10000),                               |
| (20, 'Walking Hackathon for Diabetes 2026', 'Fundraising', '2026-04-15', 25000),                      |
| (30, 'Women Diabetes Education', 'Awareness', '2026-05-10', 15000),                                   |
| (40, 'Research Fund', 'Research', '2026-02-01', 50000),                                               |
| (50, 'Walking Hackathon for Diabetes 2026', 'Fundraising', '2026-06-01', 30000);                      |
| <b>);</b>                                                                                             |

| Relational table with Values specification                                                        |
|---------------------------------------------------------------------------------------------------|
|                                                                                                   |
| <b>INSERT INTO Education_Session (SessionID, Title, SessionDate, Location, CampaignID) VALUES</b> |
| <b>(</b>                                                                                          |
| (11, 'Living with Diabetes', '2026-03-03', 'London', 10),                                         |
| (12, 'Healthy Eating for Diabetes', '2026-03-05', 'Birmingham', 10),                              |
| (13, 'Diabetes in Teenagers', '2026-05-12', 'Manchester', 30),                                    |
| (14, 'Preventing Type 2 Diabetes', '2026-05-15', 'Leeds', 30),                                    |
| (15, 'Understanding Blood Glucose', '2026-03-07', 'London', 10);                                  |
| <b>);</b>                                                                                         |

| Relational table with Values specification                                                 |
|--------------------------------------------------------------------------------------------|
|                                                                                            |
| <b>INSERT INTO Donation (DonationID, DonorID, CampaignID, Amount, DonationDate) VALUES</b> |
| <b>(</b>                                                                                   |
| (11,1, 20, 150, '2026-04-08'),                                                             |
| (22, 2, 20, 400, '2026-04-17'),                                                            |
| (33, 3, 40, 300, '2026-02-10'),                                                            |
| (44, 4, 50, 800, '2026-06-05'),                                                            |
| (55, 4, 20, 250, '2026-04-18'),                                                            |
| (66, 5, 10, 100, '2026-03-02'),                                                            |
| (77, 2, 40, 200, '2026-02-20'),                                                            |
| (88, 3, 20, 150, '2026-04-19'),                                                            |
| (99, 5, 50, 600, '2026-06-10'),                                                            |
| (100, 1, 50, 200, '2026-06-12');                                                           |
| <b>);</b>                                                                                  |

| Relational table with Values specification                                                             |
|--------------------------------------------------------------------------------------------------------|
|                                                                                                        |
| <b>INSERT INTO Research_Grant (GrantID, ProjectTitle, Institution, GrantAmount, CampaignID) VALUES</b> |
| <b>(</b>                                                                                               |
| (1, 'Glucose Monitoring', 'Hospital', 12000, 40),                                                      |
| (2, 'Lifestyle Trial', 'Centre', 18000, 40),                                                           |
| (3, 'Access in Low-Income Areas', 'Institute', 22000, 50),                                             |
| (4, 'Youth Diabetes Study', 'ClinicX', 9000, 30),                                                      |
| (5, 'Digital Coaching', 'Lab', 15000, 40);                                                             |
| <b>);</b>                                                                                              |

## 4. Querying a Database

Query 1) Who are the donors who donated more than £500?

SQL:

```
SELECT d.DonorName, SUM(o.Amount) AS TotalDonations
FROM donor d JOIN donation o ON d.DonorID = o.DonorID
GROUP BY d.DonorName
HAVING SUM(o.Amount) > 500;
```

The screenshot shows the phpMyAdmin web interface. The left sidebar displays a database structure for 'diabetes\_foundation' with tables: campaign, donation, donor, education\_session, and research\_grant. The 'donor' table is selected. The top toolbar includes options like Browse, Structure, SQL, Search, Insert, Export, Import, Privileges, Operations, and Triggers. The main area shows the SQL query: `SELECT d.DonorName, SUM(o.Amount) AS TotalDonations FROM donor d JOIN donation o ON d.DonorID = o.DonorID GROUP BY d.DonorName HAVING SUM(o.Amount) > 500;`. Below the query, a table displays the results:

| DonorName | TotalDonations |
|-----------|----------------|
| Emi Cla   | 700            |
| Jo Pat    | 1050           |
| Mo Kan    | 600            |

At the bottom, there are options to print, copy to clipboard, export, display chart, or create a view from the query results.

## Query 2) How many donations does each campaign receive? (Donations per Campaign)

SQL:

```
SELECT c.CampaignName, COUNT(o.DonationID) AS NumberOfDonations, SUM(o.Amount) AS TotalAmount
FROM campaign c LEFT JOIN donation o ON c.CampaignID = o.CampaignID
GROUP BY c.CampaignName;
```

The screenshot shows the phpMyAdmin interface for a database named 'diabetes\_foundation'. The 'campaign' table is selected. The SQL query is entered in the query box, and the results are displayed in a table format. The table has three columns: 'CampaignName', 'NumberOfDonations', and 'TotalAmount'. The results show four campaigns: 'Diabetes Awareness Week 2026' (1 donation, 100 amount), 'Research Fund' (2 donations, 500 amount), 'Walking Hackathon for Diabetes 2026' (6 donations, 2150 amount), and 'Women Diabetes Education' (1 donation, 400 amount).

| CampaignName                        | NumberOfDonations | TotalAmount |
|-------------------------------------|-------------------|-------------|
| Diabetes Awareness Week 2026        | 1                 | 100         |
| Research Fund                       | 2                 | 500         |
| Walking Hackathon for Diabetes 2026 | 6                 | 2150        |
| Women Diabetes Education            | 1                 | 400         |

### Query 3) Which education sessions will be organised under the campaign “Diabetes Awareness Week 2026”

SQL:

```
SELECT s.Title, s.SessionDate, s.Location
FROM education_session s JOIN campaign c ON s.CampaignID =
c.CampaignID
WHERE c.CampaignName = 'Diabetes Awareness Week 2026';
```

The screenshot shows the phpMyAdmin interface for the 'diabetes\_foundation' database. The 'education\_session' table is selected. The SQL query is entered in the query box and executed. The results show 3 rows of data.

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 2 (3 total, Query took 0.0020 seconds.)

```
SELECT s.Title, s.SessionDate, s.Location FROM education_session s JOIN campaign c ON s.CampaignID = c.CampaignID WHERE c.CampaignName = 'Diabetes Awareness Week 2026';
```

Options: ☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

| Title                       | SessionDate | Location   |
|-----------------------------|-------------|------------|
| Living with Diabetes        | 2026-03-03  | London     |
| Healthy Eating for Diabetes | 2026-03-05  | Birmingham |
| Understanding Blood Glucose | 2026-03-07  | London     |

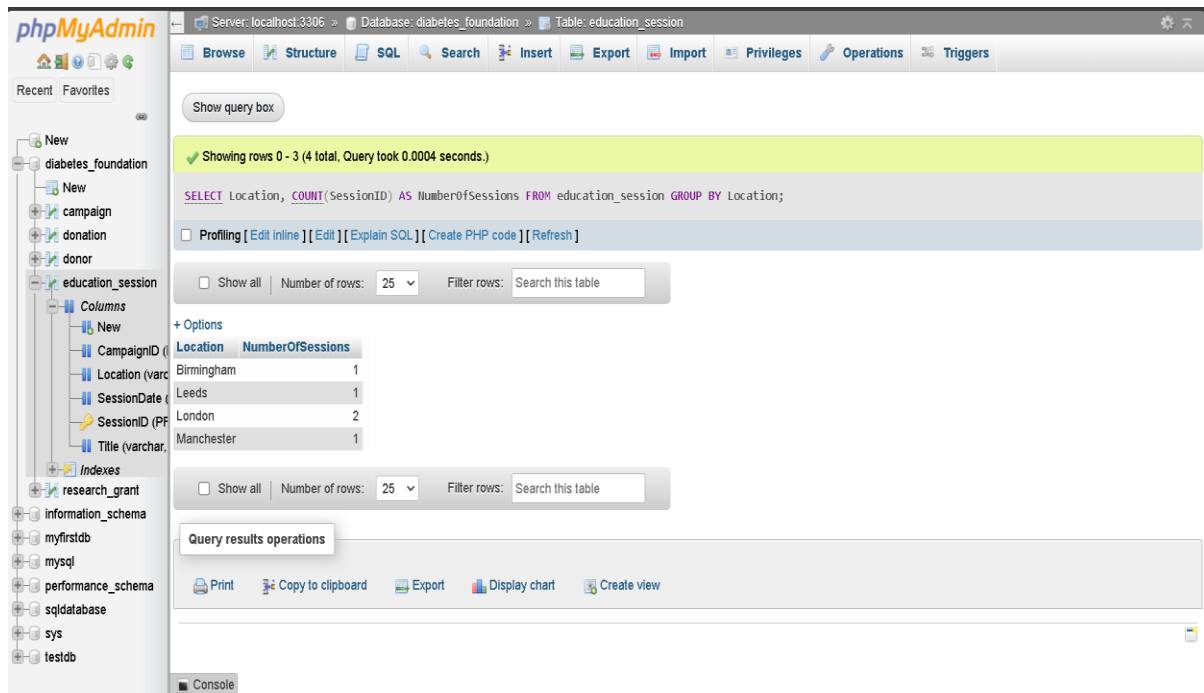
Query results operations: [Print](#) [Copy to clipboard](#) [Export](#) [Display chart](#) [Create view](#)



#### Query 4) How many education sessions will happen in each location?

SQL

```
SELECT Location, COUNT(SessionID) AS NumberOfSessions
FROM education_session
GROUP BY Location;
```



The screenshot shows the phpMyAdmin interface for a database named 'diabetes\_foundation'. The 'education\_session' table is selected. A SQL query is entered in the query box, and the results are displayed in a table. The query is: `SELECT Location, COUNT(SessionID) AS NumberOfSessions FROM education_session GROUP BY Location;`

The results table shows the following data:

| Location   | NumberOfSessions |
|------------|------------------|
| Birmingham | 1                |
| Leeds      | 1                |
| London     | 2                |
| Manchester | 1                |

Query 5) Which research projects receive grants above £10,000, and from which campaign?

SQL:

```
SELECT g.ProjectTitle, g.Institution, g.GrantAmount, c.CampaignName
FROM research_grant g JOIN campaign c ON g.CampaignID = c.CampaignID
WHERE g.GrantAmount > 10000;
```

The screenshot shows the phpMyAdmin interface for a database named 'diabetes\_foundation'. The left sidebar shows the database structure with tables: campaign, donation, donor, education\_session, research\_grant, and research\_grant. The 'research\_grant' table is selected, and its columns (CampaignID, GrantAmount, Institution, ProjectTitle) are visible. The main panel shows the SQL query and its results. The query is: `SELECT g.ProjectTitle, g.Institution, g.GrantAmount, c.CampaignName FROM research_grant g JOIN campaign c ON g.CampaignID = c.CampaignID WHERE g.GrantAmount > 10000;`. The results show 4 rows:

| ProjectTitle               | Institution | GrantAmount | CampaignName                        |
|----------------------------|-------------|-------------|-------------------------------------|
| Glucose Monitoring         | Hospital    | 12000       | Research Fund                       |
| Lifestyle Trial            | Centre      | 18000       | Research Fund                       |
| Access in Low-Income Areas | Institute   | 22000       | Walking Hackathon for Diabetes 2026 |
| Digital Coaching           | Lab         | 15000       | Research Fund                       |

## **5. Discuss the idea of storing data with a non-relational database (350 words maximum)**

If the data was stored in a non-relational database, the structure of the system will look very different from the relational version. In a relational database, everything must be in fixed tables. A NoSQL database does not need a fixed schema, so the data can grow and change more easily. This is useful, because for example in the case of the charity's dataset new types of campaigns, sessions, or research may appear over time. Also, the suitable type of NoSQL database will be a document-oriented database (such as MongoDB), and store information as JSON documents. These documents can contain different fields, nested objects, and lists.

To explain this, for example, a campaign document will include the campaign name, start date, target amount, and list of sessions inside it. Each session can have its own small document with the title, date, and location. Donations and research grants can also be stored as embedded documents. This keeps related information together.

Moreover, NoSQL databases scale horizontally, meaning we can add more servers easily when the data grows. They are also distributed across multiple locations and have built-in replication, so the system keeps running even if one server fails.

Finally, using a document-oriented NoSQL database will make the data flexible and easier to update. The JSON structure matches the data, and the scalability and reliability features make NoSQL a good choice for handling growing and changing data. An example if I will use document-oriented NoSQL database for the charity's data and create the Donor document by JSON:

```
{
  "DonorID": 1,
  "DonorName": " Ani Bro ",
  "Email": "a.b@example.org",
  "City": "London",
  "Donations": [
    {
      "DonationID": 11,
      "CampaignID": 20,
      "Amount": 150,
      "DonationDate": "2026-04-08"
    }
  ]
}
```